QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 19

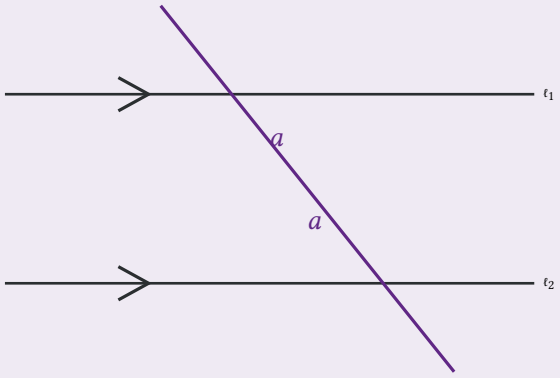
RULE · Angle relationships (burchaklar)

Type	Rule
Complementary	sum = 90°
Supplementary	sum = 180°
Vertical angles	equal

$\angle A + \angle B + \angle C = 180°$ (any triangle)

RULE · Parallel lines cut by a transversal

Pair	Rule
Corresponding angles	equal
Alternate interior angles	equal
Co-interior (same-side) angles	sum = 180°



Only applies when the lines are stated or marked (→) parallel – never assume from how a figure looks.

RULES SHEET – 2/2

RULE · Similar triangles (o'xshash)

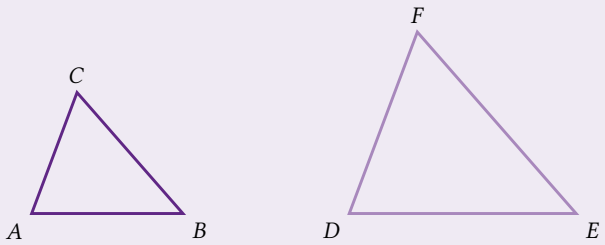
$$\triangle ABC \sim \triangle DEF \implies \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

Letter order tells you the match: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$ – proportional, not equal.

$$AB = 6, BC = 9, AC = 12, DE = 8 \implies \text{scale} = \frac{8}{6} = \frac{4}{3} \implies EF = 12,\ DF = 16$$

RULE · Proving similarity

Test	Checks
AA	two pairs of angles equal
SAS	two side pairs proportional, included angle equal
SSS	all three side pairs proportional



Same shape, different size – angles match exactly, sides scale by one constant factor.